ENERGY DISSIPATION PRESERVATION OF IMPLICIT-EXPLICIT LINEAR MULTISTEP METHODS FOR GRADIENT FLOWS II: GENERAL MULTIPLIERS

Abstract. ...

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1. Introduction. In this work, we consider the following gradient flow with periodic boundary condition:

$$u_t = \mathcal{M}\big[\mathcal{L}u + f(u)\big], \quad u(0,\boldsymbol{x}) = u^0(\boldsymbol{x}) \quad \text{for } (\boldsymbol{x},t) \in \Omega \times (0,T], \tag{1.1}$$

where $\Omega = [-\pi,\pi]^d \subset \mathbb{R}^d$ $(d = 1,2,3)$, $T > 0$ is the final time, u^0 is the initial condition, $\mathcal{M}$ is a linear negative definite operator, $\mathcal{L}$ is a linear positive definite operator, $f(u)$ represents a nonlinear but non-stiff term, and there exists a nonnegative functional $F(u)$ such that $f(u) = \frac{\delta}{\delta u}F(u)$. Then the problem (1.1) can be reformulated as

$$u_t = \mathcal{M}\frac{\delta E}{\delta u} \quad \text{with} \quad E[u] := \int_\Omega \Big[\frac{1}{2}u\mathcal{L}u + F(u)\Big]\mathrm{d}\boldsymbol{x}, \tag{1.2}$$

and the dynamics satisfies the energy dissipation law

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \Big(\frac{\delta E}{\delta u}, \frac{\partial u}{\partial t}\Big) = \Big(\mathcal{M}^{-1}\frac{\partial u}{\partial t}, \frac{\partial u}{\partial t}\Big) \le 0, \tag{1.3}$$

where $(\cdot,\cdot)$ denotes the L^2 inner product defined by $(f,g) := \int_\Omega f(\boldsymbol{x})g(\boldsymbol{x})\mathrm{d}\boldsymbol{x}$, and $\|\cdot\|$ denotes the induced L^2 norm.

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We consider a general multiplier of the form

$$\sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i} \quad \text{with} \quad \sum_{i=0}^{k-1} \mu_i = 1, \tag{1.4}$$

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2. Preliminaries. This section collects some preliminaries required for the subsequent analysis; see, e.g., [2, 6].

2.1. Abstract settings. Let H be a Hilbert space with inner product $(\cdot,\cdot)$ and induced norm $\|\cdot\|$. Assume S and V are Hilbert spaces such that $S \hookrightarrow V \hookrightarrow H$ with continuous and compact embeddings. Let V' be the dual of V and denote by $\langle\cdot,\cdot\rangle_{V',V}$ the duality pairing. Identifying H with its dual H', we have the Gelfand triple $V \subset H \subset V'$.

Assume that $\mathcal{L} : S \to V$ is a linear, self-adjoint, and positive definite operator:

$$(\mathcal{L}u, v) = (u, \mathcal{L}v), \quad (\mathcal{L}v, v) > 0, \quad \forall u, v \in S \text{ and } v \neq 0.$$

The associated semi-norm is defined by $\|\mathcal{L}^{1/2}u\| := (\mathcal{L}u, u)^{1/2}$. Assume that $\mathcal{M} : V \to V'$ is a linear, self-adjoint, and negative definite operator:

$$\langle \mathcal{M}u, v\rangle_{V',V} = \langle \mathcal{M}v, u\rangle_{V',V}, \quad \langle \mathcal{M}v, v\rangle_{V',V} < 0, \quad \forall u, v \in V \text{ and } v \neq 0.$$

Then, its inverse operator $\mathcal{M}^{-1} : V' \to V$ is still linear, self-adjoint, and negative definite. For $v \in H$, one can define the $(-\mathcal{M})^{-1}$-induced norm as $\|(-\mathcal{M})^{-1/2}v\| := (v, -\mathcal{M}^{-1}v)^{1/2}$.

Assume that $f : S \to V$ is a (possibly nonlinear) operator satisfying the global Lipschitz condition with constant $\ell_f > 0$:

$$\|f(u) - f(v)\| \leq \ell_f \|u - v\|, \quad \forall u, v \in S. \tag{2.1}$$

This implies that the potential functional F satisfies the following inequality:

$$|(F(u) - F(v), 1) - (f(v), u - v)| \leq \frac{\ell_f}{2}\|u - v\|^2, \quad \forall u, v \in S. \tag{2.2}$$

REMARK 2.1. *The Lipschitz continuity assumption on the nonlinear term is widely used in the analysis of numerical methods for gradient flows. For potentials that do not satisfy this condition globally, a common remedy in numerical simulations is to use truncation techniques, see, e.g., [9, 8, 5].*

In addition, we assume that there exist two constants $\zeta > 0$ and $0 < \eta \leq 1$ such that the following inequality holds:

$$\|v\| \leq \zeta \|(-\mathcal{M})^{-1/2}v\|^{\eta} \|\mathcal{L}^{1/2}v\|^{1-\eta}, \quad \forall v \in S. \tag{2.3}$$

We list several typical gradient flows with parameter $\varepsilon > 0$ to better understand the above notations:

- Allen–Cahn equation: $\mathcal{M} = -\mathcal{I}$, $\mathcal{L} = -\varepsilon^2\Delta + \mathcal{I}$, $f(u) = u^3 - 2u$, $H = L^2(\Omega)$, $V = L^2(\Omega)$, and $S = H^2(\Omega)$. The inequality (2.3) holds for $\zeta = 1$ and $\eta = 1$.
- Cahn–Hilliard equation: $\mathcal{M} = \Delta$, $\mathcal{L} = -\varepsilon^2\Delta$, $f(u) = u^3 - u$, $H = \dot{L}^2(\Omega)$, $V = \dot{H}^1(\Omega)$, and $S = \dot{H}^3(\Omega)$. The inequality (2.3) holds for $\zeta = \varepsilon^{-1/2}$ and $\eta = 1/2$.
- Phase field crystal equation: $\mathcal{M} = \Delta$, $\mathcal{L} = (\mathcal{I} + \Delta)^2 + \mathcal{I}$, $f(u) = u^3 - (\varepsilon + 1)u$, $H = \dot{L}^2(\Omega)$, $V = \dot{H}^1(\Omega)$, and $S = \dot{H}^5(\Omega)$. The inequality (2.3) holds for $\zeta = (2\sqrt{2} - 2)^{-1/4}$ and $\eta = 1/2$.

Here, the Sobolev spaces are actually periodic Sobolev spaces as the periodic boundary condition is equipped with, and for a Hilbert space X, $\dot{X} = \{v \in X : \int_\Omega v \mathrm{d}\boldsymbol{x} = 0\}$ denotes the subspace of functions with zero mean, which is a common requirement for mass-conserving gradient flows.

2.2. Formulation of IMEX-LMMs. Let $t_n = n\tau$, $n = 0, \ldots, N$, be a uniform partition of $[0, T]$ with step size $\tau = T/N$. Given the initial approximations $\{u^i\}_{i=0}^{k-1} \in S$, the k-step IMEX-LMM for (1.1) is

$$\sum_{i=0}^{k} A_i^{(k)} u^{n+1-i} = \tau\mathcal{M}\Big[\sum_{i=0}^{k} B_i^{(k)} \mathcal{L}u^{n+1-i} + \sum_{i=1}^{k} \hat{B}_i^{(k)} f(u^{n+1-i})\Big] \tag{2.4}$$

for $k-1 \le n \le N-1$, where the coefficients satisfy the kth-order conditions:

$$\begin{cases} \displaystyle\sum_{i=0}^{k} A_i^{(k)} = 0, \\ \displaystyle\sum_{i=0}^{k} A_i^{(k)}(-i)^{m+1} = (m+1)\sum_{i=0}^{k} B_i^{(k)}(-i)^m, \quad 0 \le m \le k-1, \\ \displaystyle\sum_{i=0}^{k} A_i^{(k)}(-i)^{m+1} = (m+1)\sum_{i=1}^{k} \hat{B}_i^{(k)}(-i)^m, \quad 0 \le m \le k-1. \end{cases} \tag{2.5}$$

To simplify the analysis, we further impose the following normalization condition:

$$\sum_{i=0}^{k} A_i^{(k)}(-i) = \sum_{i=0}^{k} B_i^{(k)} = \sum_{i=1}^{k} \hat{B}_i^{(k)} = 1, \tag{2.6}$$

which ensures that the discrete operator $\frac{1}{\tau}\sum_{i=0}^{k} A_i^{(k)} u(t_{n+1-i})$ is a first-order approximation of $u_t(t_{n+1})$. By inverting $\tau\mathcal{M}$, (2.4) can be reformulated as

$$\frac{1}{\tau}\sum_{i=0}^{k-1} a_i^{(k)} \mathcal{M}^{-1}\big(\delta u^{n+1-i}\big) - \sum_{i=0}^{k} B_i^{(k)} \mathcal{L} u^{n+1-i} = f(u^n) + \sum_{i=1}^{k-1} \hat{b}_i^{(k)} \delta f(u^{n+1-i}) \tag{2.7}$$

for $k-1 \le n \le N-1$, where $\delta u^n := u^n - u^{n-1}$ denotes the first-order time difference and the reformulated coefficients are given by the cumulative sums:

$$\begin{cases} a_i^{(k)} = \displaystyle\sum_{j=0}^{i} A_j^{(k)}, & 0 \le i \le k-1, \\ \hat{b}_i^{(k)} = \displaystyle\sum_{j=1}^{i} \hat{B}_j^{(k)} - 1, & 1 \le i \le k-1, \end{cases} \tag{2.8}$$

For simplicity, we define the coefficient vectors:

$$\boldsymbol{a}^{(k)} = [a_0^{(k)}, \dots, a_{k-1}^{(k)}]^\top, \ \boldsymbol{B}^{(k)} = [B_0^{(k)}, \dots, B_k^{(k)}]^\top, \text{ and } \hat{\boldsymbol{b}}^{(k)} = [\hat{b}_1^{(k)}, \dots, \hat{b}_{k-1}^{(k)}, 0]^\top. \tag{2.9}$$

3. Energy dissipation preservation of IMEX-LMMs. In the following analysis, we establish the energy dissipation law for the reformulation (2.7) of IMEX-LMMs using the multiplier technique. To better present our result, we rewrite (1.4) as

$$\sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i} = \sum_{i=0}^{k} \nu_i u^{n+1-i}, \tag{3.1}$$

where $\nu_0 = \mu_0$, $\nu_k = -\mu_{k-1}$, and $\nu_i = \mu_i - \mu_{i-1}$ for $1 \le i \le k-1$. We denote the corresponding coefficient vectors by

$$\boldsymbol{\mu} = [\mu_0, \dots, \mu_{k-1}]^\top \quad \text{and} \quad \boldsymbol{\nu} = [\nu_0, \dots, \nu_k]^\top. \tag{3.2}$$

Define the generating polynomial as

$$M(z; \boldsymbol{s}) = \sum_{i=0}^{r} s_i z^{r-i}$$

for a given vector $\boldsymbol{s} = [s_0, \ldots, s_r]^\top$. Here, $\boldsymbol{s}$ may stand for $\boldsymbol{a}^{(k)}$, $\boldsymbol{B}^{(k)}$, $\boldsymbol{\mu}$, or $\boldsymbol{\nu}$. By (3.1), we have $M(z;\boldsymbol{\nu}) = (z-1)M(z;\boldsymbol{\mu})$.

In addition, we shall define the following modified energy:

(3.3)

$$E_G^n = -\frac{1}{\tau}\boldsymbol{v}_n^\top G_a(\mathcal{M}^{-1}\boldsymbol{v}_n) + \boldsymbol{u}_n^\top G_B(\mathcal{L}\boldsymbol{u}_n) + \sum_{i=0}^{k-1}\mu_i(F(u^{n-i}),1) + \frac{\ell_f}{2}\sum_{i=1}^{k-1}\hat{c}_i\|\delta u^{n+1-i}\|^2$$

for two given symmetric positive definite matrices $G_a \in \mathbb{R}^{(k-1)\times(k-1)}$ and $G_B \in \mathbb{R}^{k\times k}$. Here,

$$\boldsymbol{v}_n = [\delta u^n, \delta u^{n-1}, \ldots, \delta u^{n+2-k}]^\top, \quad \boldsymbol{u}_n = [u^n, u^{n-1}, \ldots, u^{n+1-k}]^\top, \tag{3.4}$$

ℓ_f is the Lipschitz constant in (2.1), the product of two vector functions is defined by $\boldsymbol{u}^\top\boldsymbol{v} = \sum_i(u^i, v^i)$, and $\hat{c}_i = \sum_{j=i}^{k-1}\tilde{c}_j$ for $0 \le i \le k-1$, where

$$\tilde{c}_i = (1-\delta_{i,0})|\hat{b}_i^{(k)}|\sum_{j=0}^{k-1}|\mu_j| + |\mu_i|\sum_{j=1}^{k-1}|\hat{b}_j^{(k)}| + |\mu_i| + \left(|\mu_i| + \sum_{j=i}^{k-1} j|\mu_j|\right)(1-\delta_{i,0}). \tag{3.5}$$

Since G_a and G_B are symmetric positive definite, there exist constants $\lambda_a, \lambda_B > 0$ such that

$$\boldsymbol{x}^\top G_a\boldsymbol{x} \ge \lambda_a\boldsymbol{x}^\top\boldsymbol{x}, \quad \boldsymbol{y}^\top G_B\boldsymbol{y} \ge \lambda_B\boldsymbol{y}^\top\boldsymbol{y}, \quad \forall \boldsymbol{x} \in \mathbb{R}^{k-1},\ \boldsymbol{y} \in \mathbb{R}^k. \tag{3.6}$$

The modified energy (3.3) is an $\mathcal{O}(\tau)$ approximation of the original energy (1.2); see Lemma 3.5 and Remark 3.6. Under the additional assumptions of Lemma 3.4, the modified energy is uniformly nonnegative. We are now ready to state and prove our main result on the modified energy dissipation of the IMEX-LMMs for gradient flows.

THEOREM 3.1. *Assume that the gradient flow* (1.1) *has a Lipschitz continuous nonlinearity satisfying* (2.1)*, and that there exist constants* $\zeta > 0$ *and* $0 < \eta \le 1$ *such that* (2.3) *holds. Consider the* k*-step IMEX-LMM* (2.7)*, and let* $\boldsymbol{a}^{(k)}$*,* $\boldsymbol{B}^{(k)}$*, and* $\hat{\boldsymbol{b}}^{(k)}$ *be the coefficient vectors defined in* (2.9)*. Let the multiplier in* (1.4) *be specified by the vectors* $\boldsymbol{\mu}$ *and* $\boldsymbol{\nu}$ *defined in* (3.1)*.*

Suppose that the generating polynomials $M(z;\boldsymbol{a}^{(k)})$ *and* $M(z;\boldsymbol{\mu})$ *are both of degree* $k-1$*, that* $M(z;\boldsymbol{B}^{(k)})$ *is of degree* k*, that* $M(z;\boldsymbol{a}^{(k)})$ *and* $M(z;\boldsymbol{B}^{(k)})$ *are relatively prime to* $M(z;\boldsymbol{\mu})$ *and* $M(z;\boldsymbol{\nu})$*, respectively, and that* $M(z;\boldsymbol{\mu}) \ne 0$ *for* $|z| > 1$*. Assume further that there exist constants* $\alpha, \beta > 0$ *such that for any* $\theta \in [0,\pi]$*,*

$$\mathrm{Re}\Big\{M(e^{\mathrm{i}\theta};\boldsymbol{a}^{(k)})M(e^{-\mathrm{i}\theta};\boldsymbol{\mu})\Big\} \ge \alpha, \quad \mathrm{Re}\Big\{M(e^{\mathrm{i}\theta};\boldsymbol{B}^{(k)})M(e^{-\mathrm{i}\theta};\boldsymbol{\nu})\Big\} \ge 2\beta(1-\cos\theta).$$

Then there exist two symmetric positive definite matrices $G_a \in \mathbb{R}^{(k-1)\times(k-1)}$ *and* $G_B \in \mathbb{R}^{k\times k}$*, such that the modified energy* E_G^n *defined by* (3.3) *satisfies* $E_G^{n+1} \le E_G^n$ *whenever the time step* τ *satisfies*

$$0 < \tau \le \tau_{\max}, \quad \tau_{\max} := \frac{\alpha\beta^{\bar{\eta}}}{(\ell_f\hat{c}_0/2)^{1+\bar{\eta}}\eta(1-\eta)^{\bar{\eta}}\zeta^{2+2\bar{\eta}}}, \tag{3.7}$$

where ℓ_f *is the Lipschitz constant in* (2.1)*, and* $\bar{\eta} := \frac{1-\eta}{\eta}$*. Here and throughout, we adopt the convention* $0^0 = 1$*.*

Proof. The proof is deferred to subsection 3.3. ∎

3.1. Generalization of the Baiocchi–Crouzeix equivalence theorem. In this subsection, we generalize an algebraic result by Baiocchi and Crouzeix [1] (see also [3] and [6, Section. V.6]), which will be used in the proof of Theorem 3.1. Let

$$\rho(z) = \rho_0 + \rho_1 z + \ldots + \rho_r z^r \quad \text{and} \quad \sigma(z) = \sigma_0 + \sigma_1 z + \ldots + \sigma_r z^r$$

be real polynomials of degree r, where $r \geq 1$ and $\rho_r \sigma_r \neq 0$. Let $\lambda(z) = \lambda_0 + \lambda_1 z + \cdots + \lambda_r z^r$ be a nonzero real polynomial of degree at most r. For a polynomial $p(z) = \sum_{j=0}^{r} p_j z^j$, define its coefficient vector by $\boldsymbol{p} = [p_r, \ldots, p_0]^\top$. In particular, $\boldsymbol{\rho}$, $\boldsymbol{\sigma}$, and $\boldsymbol{\lambda}$ denote the coefficient vectors of ρ, σ, and λ, respectively.

PROPOSITION 3.2. *Assume that $\rho(z)$ and $\sigma(z)$ are relatively prime. Let $\gamma \geq 0$ be a constant. Then the following statements are equivalent:*

(i)

$$\operatorname{Re}\left\{\frac{\rho(z)}{\sigma(z)}\right\} > \gamma \left|\frac{\lambda(z)}{\sigma(z)}\right|^2, \quad \forall |z| > 1. \tag{3.8}$$

(ii)

$$\operatorname{Re}\{\rho(z)\sigma(\bar{z})\} \geq \gamma |\lambda(z)|^2, \quad \forall |z| \geq 1. \tag{3.9}$$

(iii) (a) $\sigma(z) \neq 0, \quad \forall |z| > 1$,
(b) $\operatorname{Re}\{\rho(z)\sigma(\bar{z})\} \geq \gamma |\lambda(z)|^2, \quad \forall |z| = 1$,
(c) *If z_0 is a root of $\sigma(z)$ with $|z_0| = 1$, then z_0 is a simple root, and* $\frac{\rho(z_0)}{\sigma'(z_0)} \bar{z}_0 > 0$.

(iv) *There exist r linearly independent polynomials $p_1, \ldots, p_r$, and q with real coefficients,* $\deg(p_j) \leq r-1$, *and* $\deg(q) \leq r$, *such that for any $z, w \in \mathbb{C}$,*

$$\rho(z)\sigma(w) + \rho(w)\sigma(z) - 2\gamma\lambda(z)\lambda(w) = q(z)q(w) + (zw-1)\sum_{j=1}^{r} p_j(z)p_j(w). \tag{3.10}$$

(v) *There exist $\boldsymbol{q} = [q_r, \ldots, q_0]^\top \in \mathbb{R}^{r+1}$ and a positive definite matrix $G \in \mathbb{R}^{r \times r}$ such that*

$$\boldsymbol{\rho}\boldsymbol{\sigma}^\top + \boldsymbol{\sigma}\boldsymbol{\rho}^\top - 2\gamma\boldsymbol{\lambda}\boldsymbol{\lambda}^\top = \boldsymbol{q}\boldsymbol{q}^\top + \begin{bmatrix} G & \\ & 0 \end{bmatrix} - \begin{bmatrix} 0 & \\ & G \end{bmatrix}. \tag{3.11}$$

Proof. We divide the proof into three parts: (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i), followed by (ii) $\Leftrightarrow$ (iv), and finally (iv) $\Leftrightarrow$ (v).

Part A: Equivalence of (i)–(iii).

(i) $\Rightarrow$ (ii): Assume that (i) holds. First, we show that $\sigma(z) \neq 0$ for all $|z| > 1$. Suppose by contradiction that $\sigma(z)$ has a root z_0 with $|z_0| > 1$ of multiplicity $m \geq 1$. Since $\rho(z)$ and $\sigma(z)$ are relatively prime, $\rho(z_0) \neq 0$. Thus, z_0 is a pole of $\rho(z)/\sigma(z)$ of order m. In a neighborhood of z_0, the Laurent expansion gives

$$\rho(z)/\sigma(z) = a_{-m}(z-z_0)^{-m} + O((z-z_0)^{-m+1}) \quad \text{with} \quad a_{-m} \neq 0. \tag{3.12}$$

Let $a_{-m} = |a_{-m}|e^{i\phi}$ and approach z_0 along the ray $z = z_0 + \varepsilon e^{i\theta}$ for $\varepsilon > 0$. The real part of the leading term in (3.12) is $|a_{-m}|\varepsilon^{-m}\cos(\phi - m\theta)$. Since $m \geq 1$, we can choose an angle θ_0 such that $\cos(\phi - m\theta_0) = -1$. Along the ray $\theta = \theta_0$, as $\varepsilon \to 0^+$,

this real part approaches $-\infty$, dominating the remaining terms. Then there exists $\varepsilon_0 > 0$ such that

$$\mathrm{Re}\,\{\rho(z)/\sigma(z)\} < 0, \quad \forall z = z_0 + \varepsilon e^{i\theta_0} \quad \text{with} \quad 0 < \varepsilon < \varepsilon_0. \tag{3.13}$$

Choosing ε small enough to ensure $|z| > 1$ contradicts condition (i). Thus, $\sigma(z) \neq 0$ for all $|z| > 1$.

Since $|\sigma(z)|^2 > 0$ for $|z| > 1$, we multiply both sides of (3.8) by $|\sigma(z)|^2 = \sigma(z)\sigma(\bar{z})$ to obtain $\mathrm{Re}\{\rho(z)\sigma(\bar{z})\} > \gamma|\lambda(z)|^2$ for all $|z| > 1$. Since the function $z \mapsto \mathrm{Re}\{\rho(z)\sigma(\bar{z})\} - \gamma|\lambda(z)|^2$ is continuous on $\mathbb{C}$, letting $|z| \to 1$ yields condition (ii).

(ii) $\Rightarrow$ (iii): Assume that (ii) holds. Condition (iii)(b) is obtained by restricting (ii) to $|z| = 1$. For condition (iii)(a), we suppose by contradiction that z_0 with $|z_0| > 1$ is a root of $\sigma(z)$ of multiplicity $m \geq 1$. By the relative primality assumption, $\rho(z_0) \neq 0$. Then by the proof of (i) $\Rightarrow$ (ii), there exists $\varepsilon_0 > 0$ and θ_0 such that (3.13) holds. Thus we have

$$\mathrm{Re}\{\rho(z)\sigma(\bar{z})\} = |\sigma(z)|^2\,\mathrm{Re}\,\{\rho(z)/\sigma(z)\} < 0, \quad \forall z = z_0 + \varepsilon e^{i\theta_0} \quad \text{with} \quad 0 < \varepsilon < \varepsilon_0.$$

By choosing small enough ε such that $|z| \geq 1$, we obtain $\mathrm{Re}\{\rho(z)\sigma(\bar{z})\} < 0$. This contradicts condition (ii).

For condition (iii)(c), assume that $z_0 = e^{i\theta_0}$ is a root of $\sigma(z)$ of multiplicity $m \geq 1$. Then z_0 is a pole of $\rho(z)/\sigma(z)$ of order m. In a neighborhood of z_0, we have the Laurent expansion (3.12) by setting $z_0 = e^{i\theta_0}$. When z approaches z_0 from the exterior $|z| > 1$ along the trajectory $z = z_0(1 + \varepsilon e^{i\theta})$ for $\varepsilon > 0$ and $\theta \in [-\pi/2, \pi/2]$, the real part of the leading term of the Laurent series of $\rho(z)/\sigma(z)$ is

$$|a_{-m}|\varepsilon^{-m}\cos(\arg(a_{-m}\bar{z}_0^m) - m\theta). \tag{3.14}$$

By condition (ii) and $\mathrm{Re}\{\rho(z)\sigma(\bar{z})\} = |\sigma(z)|^2\,\mathrm{Re}\{\rho(z)/\sigma(z)\}$, (3.14) must be nonnegative for all $\theta \in [-\pi/2, \pi/2]$ as $\varepsilon \to 0^+$. If $m \geq 2$, the angle $m\theta$ spans an interval of length at least 2π. This makes the cosine in (3.14) strictly negative for some θ, contradicting (ii). Thus, the root must be simple, i.e., $m = 1$. For $m = 1$, $\cos(\arg(a_{-1}\bar{z}_0) - \theta)$ must remain nonnegative for all $\theta \in [-\pi/2, \pi/2]$, which requires that $\arg(a_{-1}\bar{z}_0)$ is an integer multiple of 2π. Then $a_{-1}\bar{z}_0$ is a positive real number. Since the residue of $\rho(z)/\sigma(z)$ at z_0 is $a_{-1} = \rho(z_0)/\sigma'(z_0)$, this implies $\frac{\rho(z_0)}{\sigma'(z_0)}\bar{z}_0 > 0$.

(iii) $\Rightarrow$ (i): Assume that (iii) holds. Let $z_1, \ldots, z_s$ be the simple boundary roots of $\sigma(z)$ on $|z| = 1$. By condition (iii)(c), $c_j := \bar{z}_j\rho(z_j)/\sigma'(z_j) > 0$. We isolate the boundary singularities by defining

$$h(z) := \rho(z)/\sigma(z) - \sum_{j=1}^{s} c_j z_j/(z - z_j),$$

making $h(z)$ holomorphic in $|z| > 1$ and continuous for $|z| \geq 1$. Using the identity $\mathrm{Re}\{z_j/(z - z_j)\} = \frac{1}{2}(|z|^2 - 1)/|z - z_j|^2 - \frac{1}{2}$ for $z \neq z_j$, we have

$$\mathrm{Re}\left\{\frac{\rho(z)}{\sigma(z)}\right\} = \mathrm{Re}\{h(z)\} - \sum_{j=1}^{s}\frac{c_j}{2} + \sum_{j=1}^{s}\frac{c_j}{2}\frac{|z|^2 - 1}{|z - z_j|^2}. \tag{3.15}$$

Let

$$V(z) := \mathrm{Re}\left\{\frac{\rho(z)}{\sigma(z)}\right\} - \gamma\left|\frac{\lambda(z)}{\sigma(z)}\right|^2 = H(z) + \sum_{j=1}^{s}\frac{c_j}{2}\frac{|z|^2 - 1}{|z - z_j|^2}, \tag{3.16}$$

where $H(z) := \mathrm{Re}\{h(z)\} - \sum_{j=1}^{s} c_j/2 - \gamma|\lambda(z)/\sigma(z)|^2$. Since σ has real coefficients, $\bar{z}_j$ is also a root of σ. If $\gamma > 0$, then condition (iii)(b) evaluated at $z = z_j$ gives $0 = \mathrm{Re}\{\rho(z_j)\sigma(\bar{z}_j)\} \geq \gamma|\lambda(z_j)|^2$, and hence $\lambda(z_j) = 0$. Since z_j is a simple root of σ by condition (iii)(c), the quotient $\lambda(z)/\sigma(z)$ has a removable singularity at z_j. Therefore H extends continuously to $|z| = 1$. When $\gamma = 0$, the last term in H is absent, so the continuity is immediate. Thus by condition (iii)(b) and (3.16), we have $H(z) \geq 0$ for $|z| = 1$.

Using the map $w = 1/z$, which sends $|z| \geq 1$ onto $|w| \leq 1$, we define

$$\begin{aligned} &\tilde{\rho}(w) := w^r \rho(1/w),\ \tilde{\sigma}(w) := w^r \sigma(1/w),\ \tilde{\lambda}(w) := w^r \lambda(1/w), \\ &\tilde{h}(w) := h(1/w),\ \tilde{H}(w) := H(1/w). \end{aligned}$$

Then $\tilde{\rho}$, $\tilde{\sigma}$, and $\tilde{\lambda}$ are polynomials, with $\tilde{\sigma}(0) = \sigma_r \neq 0$. By the preceding discussion, $\tilde{h}$ is holomorphic in $|w| < 1$ and continuous for $|w| \leq 1$, $\tilde{\lambda}/\tilde{\sigma}$ is holomorphic in $|w| < 1$ and continuous on $|w| = 1$ when $\gamma > 0$, $\tilde{H}(w)$ is continuous for $|w| \leq 1$, and $\tilde{H}(w) \geq 0$ on $|w| = 1$.

Since $\tilde{h}(w)$ is holomorphic in $|w| < 1$ and continuous for $|w| \leq 1$, $\mathrm{Re}\{\tilde{h}(w)\}$ is harmonic for $|w| < 1$ and continuous for $|w| \leq 1$. Using the identity of the Laplacian for a holomorphic function g: $\Delta|g|^2 = 4|g'|^2$, we find $\Delta\tilde{H}(w) = -4\gamma|(\tilde{\lambda}(w)/\tilde{\sigma}(w))'|^2 \leq 0$, meaning $\tilde{H}(w)$ is a superharmonic function in $|w| < 1$ and continuous for $|w| \leq 1$. Applying the minimum principle for the superharmonic function $\tilde{H}(w)$ gives $\tilde{H}(w) \geq 0$ for $|w| < 1$. Thus $H(z) \geq 0$ for $|z| > 1$.

1) If $s \geq 1$, since $|z|^2 - 1 > 0$ for $|z| > 1$ and $c_j > 0$, by (3.16), we have

$$V(z) = H(z) + \sum_{j=1}^{s} \frac{c_j}{2} \frac{|z|^2 - 1}{|z - z_j|^2} > 0 \quad \text{for} \quad |z| > 1.$$

2) If $s = 0$, then $V(z) = H(z) \geq 0$ for $|z| > 1$. We show that $V(z) > 0$ for $|z| > 1$ by contradiction. Suppose $V(z_*) = 0$ for some $|z_*| > 1$. Then $\tilde{H}(w_*) = 0$ where $w_* = 1/z_*$ and $|w_*| < 1$. The strong minimum principle for the superharmonic function $\tilde{H}(w)$ implies $\tilde{H}(w) \equiv 0$ for $|w| < 1$, which is equivalent to $V(z) \equiv 0$ for $|z| > 1$.
   If $\gamma = 0$, we have $\mathrm{Re}\{\rho(z)/\sigma(z)\} \equiv 0$ for $|z| > 1$. The Cauchy–Riemann equations imply that $\rho(z)/\sigma(z)$ is a purely imaginary constant $\mathrm{i}\beta$. On the other hand, $\rho(x)/\sigma(x) \in \mathbb{R}$ for all sufficiently large real x, and therefore $\beta = 0$. This means $\rho(z) \equiv 0$, contradicting that ρ is a nonzero polynomial.
   If $\gamma > 0$, we have $\Delta\tilde{H}(w) = -4\gamma|(\tilde{\lambda}(w)/\tilde{\sigma}(w))'|^2 \equiv 0$ for $|w| < 1$. Thus $|(\tilde{\lambda}(w)/\tilde{\sigma}(w))'| \equiv 0$ and $\tilde{\lambda}(w)/\tilde{\sigma}(w)$ is a constant. It follows that $\mathrm{Re}\{\tilde{h}(w)\}$ is constant in $|w| < 1$ since $\tilde{H}(w) = \mathrm{Re}\{\tilde{h}(w)\} - \gamma|\tilde{\lambda}(w)/\tilde{\sigma}(w)|^2 \equiv 0$. Since $\tilde{h}$ is holomorphic, the Cauchy–Riemann equations imply that $\tilde{h}$ itself is constant. Since $s = 0$, we have $h(z) = \rho(z)/\sigma(z)$, therefore $\tilde{h}(w) = \tilde{\rho}(w)/\tilde{\sigma}(w)$. Hence there exists a constant $d \in \mathbb{C}$ such that $\tilde{\rho}(w) = d\tilde{\sigma}(w)$, equivalently, $\rho(z) = d\,\sigma(z)$. If $d = 0$, then $\rho \equiv 0$ and this is a contradiction; if $d \neq 0$, then ρ and σ are proportional, contradicting the relative primality of ρ and σ. Thus $H(z) > 0$ for $|z| > 1$.

Part B: Equivalence of (ii) *and* (iv). (iv) $\Rightarrow$ (ii): Assume that (iv) holds. Setting $w = \bar{z}$ in (iv) gives $2\,\mathrm{Re}\{\rho(z)\sigma(\bar{z})\} - 2\gamma|\lambda(z)|^2 = |q(z)|^2 + (|z|^2 - 1)\sum_{j=1}^{r} |p_j(z)|^2$. For $|z| \geq 1$, the term $|z|^2 - 1 \geq 0$, so the right-hand side is nonnegative, proving (ii).

(ii) $\Rightarrow$ (iv): Assume that (ii) holds. Define the polynomial

$$F(z, w) := \rho(z)\sigma(w) + \sigma(z)\rho(w) - 2\gamma\lambda(z)\lambda(w).$$

By (ii), $F(z,\bar{z}) \geq 0$ on $|z| = 1$. According to the Fejér–Riesz theorem (see e.g., [4, Lemma 6.1.3]), there exists a polynomial $q(z)$ with real coefficients of degree at most r such that $F(z, z^{-1}) = q(z)q(z^{-1})$ for all $|z| = 1$. Both $F(z, z^{-1})$ and $q(z)q(z^{-1})$ are Laurent polynomials in z. Since they agree on the unit circle, they agree identically for all $z \neq 0$. Define $D(z,w) := F(z,w) - q(z)q(w)$. Then $D(z, z^{-1}) \equiv 0$ for $z \neq 0$. Since $D(z,w)$ is of degree r in each variable, we have the expansion $D(z,w) = \sum_{j=0}^{r} d_j(z)w^j$, where $d_j(z)$ is the polynomial of degree at most r with real coefficients. Let

$$K(z,w) = \sum_{j=0}^{r-1} k_j(z)w^j, \quad \text{where} \quad k_j(z) = zk_{j-1}(z) - d_j(z) \quad \text{for} \quad 1 \leq j \leq r-1,$$

and the initial value $k_0(z) = -d_0(z)$. A direct comparison of the coefficients of w^j shows that $D(z,w) = (zw-1)K(z,w)$. Since $K(z,w)$ is of degree $r-1$ in each variable and $K(z,w) = K(w,z)$, there exists a real symmetric matrix $M \in \mathbb{R}^{r\times r}$ such that

$$K(z,w) = \boldsymbol{z}_{r-1}^{\top} M \boldsymbol{w}_{r-1}, \tag{3.17}$$

where $\boldsymbol{z}_{r-1} = [z^{r-1}, z^{r-2}, \ldots, 1]^{\top}$ and $\boldsymbol{w}_{r-1} = [w^{r-1}, w^{r-2}, \ldots, 1]^{\top}$. It remains to show that there exist r linearly independent polynomials $p_1, \ldots, p_r$ with real coefficients and $\deg(p_j) \leq r-1$ such that

$$K(z,w) = \sum_{j=1}^{r} p_j(z)p_j(w), \quad \forall z, w \in \mathbb{C}. \tag{3.18}$$

We proceed by splitting the remaining proof into three steps.

Step 1: Show that M has the form (3.24). Define $A(z) = (\rho(z) + \sigma(z))/\sqrt{2}$, $B(z) = (\rho(z) - \sigma(z))/\sqrt{2}$, and $C(z) = \sqrt{2\gamma}\lambda(z)$, so $F(z,w) = A(z)A(w) - B(z)B(w) - C(z)C(w)$. Denote $A(z) = \boldsymbol{z}_r^{\top}\boldsymbol{a}$, $B(z) = \boldsymbol{z}_r^{\top}\boldsymbol{b}$, $C(z) = \boldsymbol{z}_r^{\top}\boldsymbol{c}$, and $q(z) = \boldsymbol{z}_r^{\top}\boldsymbol{q}$, respectively, where $\boldsymbol{a} = [a_r, a_{r-1}, \ldots, a_0]^{\top}$, $\boldsymbol{b} = [b_r, b_{r-1}, \ldots, b_0]^{\top}$, $\boldsymbol{c} = \sqrt{2\gamma}\boldsymbol{\lambda}$, and $\boldsymbol{q} = [q_r, q_{r-1}, \ldots, q_0]^{\top}$. Then

$$D(z,w) = \boldsymbol{z}_r^{\top}\boldsymbol{a}\boldsymbol{a}^{\top}\boldsymbol{w}_r - \boldsymbol{z}_r^{\top}\boldsymbol{b}\boldsymbol{b}^{\top}\boldsymbol{w}_r - \boldsymbol{z}_r^{\top}\boldsymbol{c}\boldsymbol{c}^{\top}\boldsymbol{w}_r - \boldsymbol{z}_r^{\top}\boldsymbol{q}\boldsymbol{q}^{\top}\boldsymbol{w}_r. \tag{3.19}$$

Equating coefficients of $D(z,w) = (zw-1)K(z,w)$ via (3.17) and (3.19) gives

$$\boldsymbol{a}\boldsymbol{a}^{\top} - \boldsymbol{b}\boldsymbol{b}^{\top} - \boldsymbol{c}\boldsymbol{c}^{\top} - \boldsymbol{q}\boldsymbol{q}^{\top} = P_0^{\top} M P_0 - S_0^{\top} M S_0, \tag{3.20}$$

where $P_0 = [I_r \;\; \boldsymbol{0}_{r\times 1}]$, $S_0 = [\boldsymbol{0}_{r\times 1} \;\; I_r]$, and $I_r \in \mathbb{R}^{r\times r}$ is the identity matrix.

The leading coefficient of $A(z)$ is $a_r = (\rho_r + \sigma_r)/\sqrt{2}$. If $a_r = 0$, then $\sigma_r = -\rho_r \neq 0$. As $x \to \infty$ on the real line, the leading term of $F(x,x)$ is $2\rho_r\sigma_r x^{2r} - 2\gamma\lambda_r^2 x^{2r} = -2(\rho_r^2 + \gamma\lambda_r^2)x^{2r} < 0$, which contradicts $F(x,x) \geq 0$ for $x \geq 1$. Thus, $a_r \neq 0$, meaning $A(z)$ has degree r. Moreover, condition (ii) implies $|A(z)|^2 \geq |B(z)|^2 + |C(z)|^2$ for all $|z| \geq 1$. If $A(z_0) = 0$ for some $|z_0| \geq 1$, then $|B(z_0)|^2 + |C(z_0)|^2 \leq |A(z_0)|^2 = 0$, and hence $B(z_0) = C(z_0) = 0$. In particular, $\rho(z_0) = \sigma(z_0) = 0$, contradicting the relative primality of ρ and σ. Thus, all roots of $A(z)$ lie strictly inside the open unit disk. Define the matrix

$$L = \begin{bmatrix} -a_{r-1}/a_r & \cdots & -a_0/a_r \\ & I_r & \end{bmatrix} \in \mathbb{R}^{(r+1)\times r}. \tag{3.21}$$

Then $\boldsymbol{a}^\top L = \mathbf{0}^\top$, $S_0 L = I_r$, and $P_0 L = J$, where

$$J = \begin{bmatrix} -\frac{a_{r-1}}{a_r} & -\frac{a_{r-2}}{a_r} & \cdots & -\frac{a_1}{a_r} & -\frac{a_0}{a_r} \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \in \mathbb{R}^{r\times r}.$$

Then the eigenvalues of J are precisely the roots of $A(z)$, counted with algebraic multiplicity. Then the spectral radius of J is strictly less than 1.

Multiplying (3.20) from the left by $L^\top$ and right by L yields the matrix equation:

$$M - J^\top M J = Q, \quad \text{where} \quad Q := L^\top(\boldsymbol{b}\boldsymbol{b}^\top + \boldsymbol{c}\boldsymbol{c}^\top + \boldsymbol{q}\boldsymbol{q}^\top)L. \tag{3.22}$$

Let $\mathrm{vec} : \mathbb{R}^{r\times r} \to \mathbb{R}^{r^2}$ be the operator from a matrix to a column vector by stacking its columns, and let $\otimes$ denote the Kronecker product. The eigenvalues of $J^\top \otimes J^\top$ are the product of any two eigenvalues of J, all of which have modulus strictly less than 1. Then $I_{r^2} - J^\top \otimes J^\top$ is nonsingular. Taking vec on both sides of (3.22) and applying the property: $\mathrm{vec}(AXB) = (B^\top \otimes A)\,\mathrm{vec}(X)$, we have

$$\mathrm{vec}(M) = (I_{r^2} - J^\top \otimes J^\top)^{-1}\,\mathrm{vec}(Q). \tag{3.23}$$

By [7, Lemma 5.6.10 and Corollary 5.6.16], we have the formula

$$(I_{r^2} - J^\top \otimes J^\top)^{-1} = \sum_{m=0}^{\infty} (J^\top \otimes J^\top)^m.$$

Applying the property $(B \otimes C)^m = B^m \otimes C^m$ to (3.23), we obtain

$$M = \sum_{m=0}^{\infty} (J^\top)^m Q J^m. \tag{3.24}$$

Step 2: Show that M is positive definite. (3.24) shows that M is positive semi-definite. We proceed by contradiction and assume that M is not strictly positive definite. Then there must exist a nonzero real vector $\boldsymbol{x} \in \mathbb{R}^r$ such that $\boldsymbol{x}^\top M \boldsymbol{x} = 0$. Then

$$\sum_{m=0}^{\infty} \boldsymbol{x}^\top (J^\top)^m Q J^m \boldsymbol{x} = 0.$$

Since Q is positive semi-definite and each term in the sum is a nonnegative real number, the infinite sum evaluating to exactly zero requires every individual term to vanish. Consequently, $\boldsymbol{x}^\top (J^\top)^m Q J^m \boldsymbol{x} = 0$ for $m \geq 0$. By the definition of Q in (3.22), we have $\boldsymbol{b}^\top L J^m \boldsymbol{x} = 0$ for $m \geq 0$. Then $\boldsymbol{x} \in \mathcal{N}$, where $\mathcal{N}$ is the null space

$$\mathcal{N} = \{\boldsymbol{y} \in \mathbb{C}^r \mid \boldsymbol{b}^\top L J^m \boldsymbol{y} = 0 \text{ for all } m \geq 0\}. \tag{3.25}$$

Since $\boldsymbol{x} \neq \mathbf{0}$, the subspace $\mathcal{N}$ is a non-trivial subspace of $\mathbb{C}^r$. By definition, $\mathcal{N}$ is an invariant subspace of the matrix J, since $\boldsymbol{y} \in \mathcal{N}$ implies $J\boldsymbol{y} \in \mathcal{N}$. Then the restriction

$J|_{\mathcal{N}}$ has an eigenpair $\lambda_{\text{eig}} \in \mathbb{C}$ and $\boldsymbol{v}$ with $\mathbf{0} \neq \boldsymbol{v} \in \mathcal{N}$. Let the vector $\boldsymbol{u} := L\boldsymbol{v} \in \mathbb{C}^{r+1}$. Applying P_0 and S_0 to $\boldsymbol{u}$ yields

$$P_0\boldsymbol{u} = P_0 L\boldsymbol{v} = J\boldsymbol{v} = \lambda_{\text{eig}}\boldsymbol{v} \quad \text{and} \quad S_0\boldsymbol{u} = S_0 L\boldsymbol{v} = \boldsymbol{v}.$$

Then $P_0\boldsymbol{u} = \lambda_{\text{eig}} S_0\boldsymbol{u}$. Let us suppose $\boldsymbol{u} = [u_r, u_{r-1}, \ldots, u_0]^\top$. The relation $P_0\boldsymbol{u} = \lambda_{\text{eig}} S_0\boldsymbol{u}$ implies that

$$u_j = \lambda_{\text{eig}}^j u_0 \quad \text{for} \quad j = 1, \ldots, r.$$

Then $\boldsymbol{u} = u_0 \boldsymbol{z}_r(\lambda_{\text{eig}})$, where $\boldsymbol{z}_r(\lambda_{\text{eig}}) = [\lambda_{\text{eig}}^r, \lambda_{\text{eig}}^{r-1}, \ldots, 1]^\top$. Since $\boldsymbol{v} = S_0\boldsymbol{u} \neq \mathbf{0}$, then $u_0 \neq 0$. Without loss of generality, we can scale the eigenvector $\boldsymbol{v}$ such that $u_0 = 1$, yielding

$$L\boldsymbol{v} = \boldsymbol{u} = \boldsymbol{z}_r(\lambda_{\text{eig}}).$$

Then on the one hand, by utilizing the property that $\boldsymbol{a}^\top L = \mathbf{0}^\top$, we have

$$0 = \boldsymbol{a}^\top (L\boldsymbol{v}) = \boldsymbol{a}^\top \boldsymbol{z}_r(\lambda_{\text{eig}}) = A(\lambda_{\text{eig}}).$$

On the other hand, since $\boldsymbol{v} \in \mathcal{N}$, then $\boldsymbol{b}^\top L\boldsymbol{v} = 0$ and

$$0 = \boldsymbol{b}^\top (L\boldsymbol{v}) = \boldsymbol{b}^\top \boldsymbol{z}_r(\lambda_{\text{eig}}) = B(\lambda_{\text{eig}}).$$

Then the complex number λ_{eig} is a common root of both polynomials $A(z)$ and $B(z)$. We have $\rho(\lambda_{\text{eig}}) = [A(\lambda_{\text{eig}}) + B(\lambda_{\text{eig}})]/\sqrt{2} = 0$ and $\sigma(\lambda_{\text{eig}}) = [A(\lambda_{\text{eig}}) - B(\lambda_{\text{eig}})]/\sqrt{2} = 0$. That is, $\rho(z)$ and $\sigma(z)$ share a common root $z = \lambda_{\text{eig}} \in \mathbb{C}$. This contradicts the assumption that the polynomials $\rho(z)$ and $\sigma(z)$ are relatively prime.

Step 3: Show that (3.18) *holds.*

Since M is real symmetric positive definite, let $M = PP^\top$ be its Cholesky factorization, where $P = [\boldsymbol{p}_1, \ldots, \boldsymbol{p}_r] \in \mathbb{R}^{r\times r}$. Define $p_j(z) = \boldsymbol{z}_{r-1}^\top \boldsymbol{p}_j$. Then each p_j has real coefficients and degree at most $r-1$, and

$$K(z,w) = \boldsymbol{z}_{r-1}^\top M \boldsymbol{w}_{r-1} = \sum_{j=1}^{r} p_j(z) p_j(w).$$

Because P is nonsingular, the polynomials $p_1, \ldots, p_r$ are linearly independent.

Part C: Equivalence of (iv) *and* (v). Let $\boldsymbol{p}_j \in \mathbb{R}^r$ be the corresponding coefficient vector of p_j, and set $P = [\boldsymbol{p}_1, \ldots, \boldsymbol{p}_r] \in \mathbb{R}^{r\times r}$ and $G = PP^\top$. Then $\sum_{j=1}^r p_j(z)p_j(w) = \boldsymbol{z}_{r-1}^\top G\, \boldsymbol{w}_{r-1}$. Therefore, condition (iv) is equivalent to

$$\boldsymbol{z}_r^\top (\boldsymbol{\rho}\boldsymbol{\sigma}^\top + \boldsymbol{\sigma}\boldsymbol{\rho}^\top - 2\gamma\boldsymbol{\lambda}\boldsymbol{\lambda}^\top)\boldsymbol{w}_r = \boldsymbol{z}_r^\top \boldsymbol{q}\boldsymbol{q}^\top \boldsymbol{w}_r + (zw-1)\boldsymbol{z}_{r-1}^\top G\, \boldsymbol{w}_{r-1}, \tag{3.26}$$

Note that

$$\boldsymbol{z}_r^\top \begin{bmatrix} G & \\ & 0 \end{bmatrix} \boldsymbol{w}_r = zw\, \boldsymbol{z}_{r-1}^\top G\, \boldsymbol{w}_{r-1} \quad \text{and} \quad \boldsymbol{z}_r^\top \begin{bmatrix} 0 & \\ & G \end{bmatrix} \boldsymbol{w}_r = \boldsymbol{z}_{r-1}^\top G\, \boldsymbol{w}_{r-1}. \tag{3.27}$$

Substituting (3.27) into (3.26) and equating the coefficients give (3.11). Moreover, the polynomials $p_1, \ldots, p_r$ are linearly independent if and only if the vectors $\boldsymbol{p}_1, \ldots, \boldsymbol{p}_r$ are linearly independent, i.e., if and only if P is nonsingular, which is equivalent to $G = PP^\top$ being positive definite. ∎

3.2. Properties of the modified energy. In this subsection, we establish three properties of the modified energy E_G^n: dissipation, non-negativity, and consistency. First, Lemma 3.3 gives sufficient positive-definiteness conditions for the energy dissipation law of IMEX-LMMs. Next, Lemma 3.4 proves the uniform non-negativity of E_G^n. Finally, Lemma 3.5 and Remark 3.6 establish its consistency with the continuous energy. Henceforth, $\mathbf{0}_{m\times n}$ denotes the zero matrix in $\mathbb{R}^{m\times n}$; whenever the dimension is clear from the context, the subscripts are omitted.

LEMMA 3.3 (Dissipation). *Assume that the gradient flow* (1.1) *has a Lipschitz continuous nonlinearity satisfying* (2.1)*, and that there exist constants* $\zeta > 0$ *and* $0 < \eta \le 1$ *such that* (2.3) *holds. Consider the* k*-step IMEX-LMM* (2.7)*, and let* $\boldsymbol{a}^{(k)}$*,* $\boldsymbol{B}^{(k)}$*, and* $\hat{\boldsymbol{b}}^{(k)}$ *be the coefficient vectors defined in* (2.9)*. Let the multiplier in* (1.4) *be specified by the vectors* $\boldsymbol{\mu}$ *and* $\boldsymbol{\nu}$ *defined in* (3.1)*.*

Suppose that there exist two symmetric positive definite matrices $G_a \in \mathbb{R}^{(k-1)\times(k-1)}$ *and* $G_B \in \mathbb{R}^{k\times k}$*, and* $\alpha, \beta > 0$ *satisfying the following positive-definiteness conditions:*

$$\boldsymbol{x}^\top U_a \boldsymbol{x} \ge \alpha x_1^2, \quad \boldsymbol{y}^\top U_B \boldsymbol{y} \ge \beta (y_1 - y_2)^2 \tag{3.28}$$

for any $\boldsymbol{x} = [x_1, \dots, x_k]^\top \in \mathbb{R}^k$ *and* $\boldsymbol{y} = [y_1, y_2, \dots, y_{k+1}]^\top \in \mathbb{R}^{k+1}$*, where*

$$U_a = \frac{1}{2}\big(\boldsymbol{\mu}(\boldsymbol{a}^{(k)})^\top + \boldsymbol{a}^{(k)}\boldsymbol{\mu}^\top\big) - \begin{bmatrix} G_a & \mathbf{0} \\ \mathbf{0}^\top & 0 \end{bmatrix} + \begin{bmatrix} 0 & \mathbf{0}^\top \\ \mathbf{0} & G_a \end{bmatrix}, \tag{3.29}$$

$$U_B = \frac{1}{2}\big(\boldsymbol{\nu}(\boldsymbol{B}^{(k)})^\top + \boldsymbol{B}^{(k)}\boldsymbol{\nu}^\top\big) - \begin{bmatrix} G_B & \mathbf{0} \\ \mathbf{0}^\top & 0 \end{bmatrix} + \begin{bmatrix} 0 & \mathbf{0}^\top \\ \mathbf{0} & G_B \end{bmatrix}. \tag{3.30}$$

Then the modified energy E_G^n *defined in* (3.3) *satisfies* $E_G^{n+1} \le E_G^n$ *under the time-step restriction* (3.7)*.*

Proof. Taking the inner product of (2.7) with $\sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i}$, we obtain

$$I_1 + I_2 = I_3 + I_4, \tag{3.31}$$

where

$$\begin{aligned} I_1 &= \frac{1}{\tau}\left(\sum_{i=0}^{k-1} a_i^{(k)} \mathcal{M}^{-1}\delta u^{n+1-i}, \sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i}\right), \\ I_2 &= -\left(\sum_{i=0}^{k} B_i^{(k)} \mathcal{L} u^{n+1-i}, \sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i}\right) = -\left(\sum_{i=0}^{k} B_i^{(k)} \mathcal{L} u^{n+1-i}, \sum_{i=0}^{k} \nu_i u^{n+1-i}\right), \\ I_3 &= \left(f(u^n), \sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i}\right), \ I_4 = \left(\sum_{i=1}^{k-1} \hat{b}_i^{(k)} \delta f(u^{n+1-i}), \sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i}\right). \end{aligned} \tag{3.31}$$

Let $\boldsymbol{w}_n = [\delta u^n, \delta u^{n-1}, \dots, \delta u^{n+1-k}]^\top$ and $\tilde{\boldsymbol{w}}_n = [u^n, u^{n-1}, \dots, u^{n-k}]^\top$. By multiplying (3.30) from the left by $\tilde{\boldsymbol{w}}_{n+1}^\top$, and from the right by $\mathcal{L}\tilde{\boldsymbol{w}}_{n+1}$, we obtain

$$I_2 = -\tilde{\boldsymbol{w}}_{n+1}^\top U_B(\mathcal{L}\tilde{\boldsymbol{w}}_{n+1}) - \boldsymbol{u}_{n+1}^\top G_B(\mathcal{L}\boldsymbol{u}_{n+1}) + \boldsymbol{u}_n^\top G_B(\mathcal{L}\boldsymbol{u}_n), \tag{3.32}$$

Analogously, I_1 can be rewritten by using (3.29) as

$$I_1 = \frac{1}{\tau}\boldsymbol{w}_{n+1}^\top U_a(\mathcal{M}^{-1}\boldsymbol{w}_{n+1}) + \frac{1}{\tau}\boldsymbol{v}_{n+1}^\top G_a(\mathcal{M}^{-1}\boldsymbol{v}_{n+1}) - \frac{1}{\tau}\boldsymbol{v}_n^\top G_a(\mathcal{M}^{-1}\boldsymbol{v}_n). \tag{3.33}$$

We split I_3 into two parts:

$$I_3 = \left(f(u^n), \sum_{i=0}^{k-1} \mu_i \delta u^{n+1-i} \right) = \sum_{i=0}^{k-1} \mu_i (f(u^n), u^{n+1-i} - u^{n-i}) = J_1 + J_2, \tag{3.34}$$

where

$$J_1 = \sum_{i=1}^{k-1} \mu_i (f(u^n) - f(u^{n-i}), u^{n+1-i} - u^{n-i}), \;\; J_2 = \sum_{i=0}^{k-1} \mu_i (f(u^{n-i}), u^{n+1-i} - u^{n-i}).$$

By (2.1), the Cauchy–Schwarz inequality, and $2ab \le a^2 + b^2$, we have

$$\begin{aligned} J_1 &\ge -\sum_{i=1}^{k-1} |\mu_i| \|f(u^n) - f(u^{n-i})\| \|\delta u^{n+1-i}\| \ge -\ell_f \sum_{i=1}^{k-1} |\mu_i| \|u^n - u^{n-i}\| \|\delta u^{n+1-i}\| \\ &\ge -\frac{\ell_f}{2} \sum_{i=1}^{k-1} |\mu_i| \left(\|u^n - u^{n-i}\|^2 + \|\delta u^{n+1-i}\|^2 \right) \\ &\ge -\frac{\ell_f}{2} \sum_{i=1}^{k-1} |\mu_i| \left(i \sum_{j=1}^{i} \left\|\delta u^{n+1-j}\right\|^2 + \|\delta u^{n+1-i}\|^2 \right) \\ &= -\frac{\ell_f}{2} \sum_{i=1}^{k-1} \left(|\mu_i| + \sum_{j=i}^{k-1} j |\mu_j| \right) \|\delta u^{n+1-i}\|^2, \end{aligned} \tag{3.35}$$

where in the last inequality we used the identity $u^n - u^{n-i} = \sum_{j=1}^{i} \delta u^{n+1-j}$ and the inequality $\|\sum_{j=1}^{i} x_j\|^2 \le i \sum_{j=1}^{i} \|x_j\|^2$. To estimate J_2, we apply (2.2) with $u = u^{n+1-i}$ and $v = u^{n-i}$ to obtain

$$|(F(u^{n+1-i}) - F(u^{n-i}), 1) - (f(u^{n-i}), u^{n+1-i} - u^{n-i})| \le \frac{\ell_f}{2} \|u^{n+1-i} - u^{n-i}\|^2. \tag{3.36}$$

Then

$$\mu_i (f(u^{n-i}), u^{n+1-i} - u^{n-i}) \ge \mu_i (F(u^{n+1-i}) - F(u^{n-i}), 1) - \frac{\ell_f}{2} |\mu_i| \|u^{n+1-i} - u^{n-i}\|^2. \tag{3.37}$$

Summing (3.37) over i gives

$$J_2 \ge \sum_{i=0}^{k-1} \mu_i (F(u^{n+1-i}) - F(u^{n-i}), 1) - \frac{\ell_f}{2} \sum_{i=0}^{k-1} |\mu_i| \|u^{n+1-i} - u^{n-i}\|^2. \tag{3.38}$$

Substituting (3.35) and (3.38) into (3.34) gives

$$\begin{aligned} I_3 \ge &-\frac{\ell_f}{2} \sum_{i=1}^{k-1} \left(|\mu_i| + \sum_{j=i}^{k-1} j |\mu_j| \right) \|\delta u^{n+1-i}\|^2 \\ &+ \sum_{i=0}^{k-1} \mu_i (F(u^{n+1-i}) - F(u^{n-i}), 1) - \frac{\ell_f}{2} \sum_{i=0}^{k-1} |\mu_i| \|u^{n+1-i} - u^{n-i}\|^2. \end{aligned} \tag{3.39}$$

For I_4, by (2.1), the Cauchy–Schwarz inequality, and $2ab \le a^2 + b^2$, we obtain

$$I_4 \ge -\frac{\ell_f}{2}\sum_{i=0}^{k-1}\Big((1-\delta_{i0})|\hat{b}_i^{(k)}|\sum_{j=0}^{k-1}|\mu_j| + |\mu_i|\sum_{j=1}^{k-1}|\hat{b}_j^{(k)}|\Big)\|\delta u^{n+1-i}\|^2. \tag{3.40}$$

Substituting (3.32), (3.33), (3.39) and (3.40) into (3.31), we have

$$\begin{aligned}
&-\frac{1}{\tau}\boldsymbol{v}_{n+1}^\top G_a(\mathcal{M}^{-1}\boldsymbol{v}_{n+1}) + \boldsymbol{u}_{n+1}^\top G_B(\mathcal{L}\boldsymbol{u}_{n+1}) \\
&+\frac{1}{\tau}\boldsymbol{v}_n^\top G_a(\mathcal{M}^{-1}\boldsymbol{v}_n) - \boldsymbol{u}_n^\top G_B(\mathcal{L}\boldsymbol{u}_n) + \sum_{i=0}^{k-1}\mu_i(F(u^{n+1-i}) - F(u^{n-i}), 1) \\
&\le \frac{1}{\tau}\boldsymbol{w}_{n+1}^\top U_a(\mathcal{M}^{-1}\boldsymbol{w}_{n+1}) - \tilde{\boldsymbol{w}}_{n+1}^\top U_B(\mathcal{L}\tilde{\boldsymbol{w}}_{n+1}) + \frac{\ell_f}{2}\sum_{i=0}^{k-1}\tilde{c}_i\|\delta u^{n+1-i}\|^2,
\end{aligned} \tag{3.41}$$

where $\tilde{c}_i$ is defined in (3.5) by collecting the coefficients of $\|\delta u^{n+1-i}\|^2$ in the estimates for I_3 and I_4. We rewrite

$$\begin{aligned}
\sum_{i=0}^{k-1}\tilde{c}_i\|\delta u^{n+1-i}\|^2 &= \sum_{i=1}^{k-1}\tilde{c}_i\|\delta u^{n+1-i}\|^2 + \tilde{c}_0\|\delta u^{n+1}\|^2 \\
&= \sum_{i=1}^{k-1}\hat{c}_i\|\delta u^{n+1-i}\|^2 - \sum_{i=1}^{k-1}\hat{c}_i\|\delta u^{n+2-i}\|^2 + \hat{c}_0\|\delta u^{n+1}\|^2,
\end{aligned} \tag{3.42}$$

where $\hat{c}_i = \sum_{j=i}^{k-1}\tilde{c}_j$ for $0 \le i \le k-1$. Substituting (3.42) into (3.41), and using the definition of E_G^n in (3.3), we have

$$\begin{aligned}
&E_G^{n+1} - E_G^n \\
&\le \frac{1}{\tau}\boldsymbol{w}_{n+1}^\top U_a(\mathcal{M}^{-1}\boldsymbol{w}_{n+1}) - \tilde{\boldsymbol{w}}_{n+1}^\top U_B(\mathcal{L}\tilde{\boldsymbol{w}}_{n+1}) + \frac{\ell_f\hat{c}_0}{2}\|\delta u^{n+1}\|^2 \\
&\le -\frac{\alpha}{\tau}\left\|(-\mathcal{M})^{-1/2}\delta u^{n+1}\right\|^2 - \beta\left\|\mathcal{L}^{1/2}\delta u^{n+1}\right\|^2 + \frac{\ell_f\hat{c}_0}{2}\|\delta u^{n+1}\|^2.
\end{aligned} \tag{3.43}$$

In the case of $\eta = 1$, (2.3) gives $\|\delta u^{n+1}\| \le \zeta\|(-\mathcal{M})^{-1/2}\delta u^{n+1}\|$. It is straightforward to verify that $E_G^{n+1} \le E_G^n$ when τ satisfies (3.7). In the case of $0 < \eta < 1$, we apply Young's inequality $ab \le \frac{a^p}{p} + \frac{b^q}{q}$ with $p = \frac{1}{\eta}$ and $q = \frac{1}{1-\eta}$. Replacing a and b by $\xi^{\eta-1}a^{2\eta}$ and $\xi^{1-\eta}b^{2-2\eta}$ respectively for any $\xi > 0$, yields

$$a^{2\eta}b^{2-2\eta} \le \eta\xi^{-\frac{1-\eta}{\eta}}a^2 + (1-\eta)\xi b^2. \tag{3.44}$$

Applying (3.44) to (2.3), we have

$$\begin{aligned}
\|\delta u^{n+1}\|^2 &\le \zeta^2\|(-\mathcal{M})^{-1/2}\delta u^{n+1}\|^{2\eta}\|\mathcal{L}^{1/2}\delta u^{n+1}\|^{2-2\eta} \\
&\le \zeta^2\left[\eta\xi^{-\frac{1-\eta}{\eta}}\|(-\mathcal{M})^{-1/2}\delta u^{n+1}\|^2 + (1-\eta)\xi\|\mathcal{L}^{1/2}\delta u^{n+1}\|^2\right].
\end{aligned} \tag{3.45}$$

To ensure the decay of E_G^n in (3.43), we then impose

$$-\frac{\alpha}{\tau} + \frac{\ell_f\hat{c}_0}{2}\zeta^2\eta\xi^{-\frac{1-\eta}{\eta}} \le 0 \text{ and } -\beta + \frac{\ell_f\hat{c}_0}{2}\zeta^2(1-\eta)\xi \le 0. \tag{3.46}$$

By taking

$$\xi = \frac{\beta}{\frac{\ell_f \hat{c}_0}{2}\zeta^2(1-\eta)},$$

the second inequality of (3.46) holds naturally and we then obtain the time-step restriction (3.7). ∎

The following lemma shows the non-negativity of E_G^n.

LEMMA 3.4 (Non-negativity). *Suppose that the assumptions of Lemma 3.3 hold. Let $\hat{c}_i = \sum_{j=i}^{k-1} \tilde{c}_j$ for $0 \le i \le k-1$, with $\tilde{c}_j$ defined in (3.5), and define the constants*

$$\bar{c}_i = \sum_{q=0}^{k-1} |\mu_q| \sum_{m=0}^{i-1} \sum_{j=i}^{k-1} |\mu_m|\,|\mu_j|\,(j-m). \tag{3.47}$$

Then, the modified energy E_G^n defined in (3.3) satisfies $E_G^n \ge 0$ uniformly, provided that either of the following two conditions holds:

(i) *$\hat{c}_i \ge 2\bar{c}_i$ for all $i = 1, \dots, k-1$;*

(ii) *$\hat{c}_i < 2\bar{c}_i$ for some $1 \le i \le k-1$, and the time-step size satisfies the restriction:*

$$0 < \tau \le \bar{\tau}_{\max}, \quad \bar{\tau}_{\max} := \frac{\lambda_a 4^{-\bar{\eta}} \lambda_B^{\bar{\eta}}}{[\ell_f(\bar{c}_{\max} - \hat{c}_{\min}/2)]^{1+\bar{\eta}} \eta (1-\eta)^{\bar{\eta}} \zeta^{2+2\bar{\eta}}}, \tag{3.48}$$

with λ_a and λ_B defined in (3.6), $\bar{\eta} = \frac{1-\eta}{\eta}$, $\hat{c}_{\min} = \min_{1\le i\le k-1} \hat{c}_i$, and $\bar{c}_{\max} = \max_{1\le i\le k-1} \bar{c}_i$.

Proof. Let $\bar{u} = \sum_{j=0}^{k-1} \mu_j u^{n-j}$. Applying (2.2) with $u = u^{n-i}$ and $v = \bar{u}$, we have

$$|(F(u^{n-i}) - F(\bar{u}), 1) - (f(\bar{u}), u^{n-i} - \bar{u})| \le \frac{\ell_f}{2}\|u^{n-i} - \bar{u}\|^2. \tag{3.49}$$

Then

$$\mu_i(F(u^{n-i}), 1) \ge \mu_i(F(\bar{u}), 1) + \mu_i(f(\bar{u}), u^{n-i} - \bar{u}) - |\mu_i|\frac{\ell_f}{2}\|u^{n-i} - \bar{u}\|^2. \tag{3.50}$$

Summing (3.50) over i gives

$$\begin{aligned}\sum_{i=0}^{k-1} \mu_i(F(u^{n-i}), 1) &\ge \sum_{i=0}^{k-1} \left[\mu_i(F(\bar{u}), 1) + \mu_i(f(\bar{u}), u^{n-i} - \bar{u})\right] - \frac{\ell_f}{2}\sum_{i=0}^{k-1} |\mu_i|\|u^{n-i} - \bar{u}\|^2 \\ &\ge -\frac{\ell_f}{2}\sum_{i=0}^{k-1} |\mu_i|\|u^{n-i} - \bar{u}\|^2,\end{aligned} \tag{3.51}$$

where we use the fact that $\sum_{i=0}^{k-1} \mu_i = 1$, $(F(\bar{u}), 1) \ge 0$, and $\sum_{i=0}^{k-1} \mu_i(u^{n-i} - \bar{u}) = 0$. By the Cauchy–Schwarz inequality, we have

$$\|u^{n-i} - \bar{u}\|^2 = \left\|\sum_{j=0}^{k-1} \mu_j \left(u^{n-i} - u^{n-j}\right)\right\|^2 \le \sum_{q=0}^{k-1} |\mu_q| \sum_{j=0}^{k-1} |\mu_j|\,\|u^{n-i} - u^{n-j}\|^2. \tag{3.52}$$

Then

$$
\begin{aligned}
-\frac{\ell_f}{2}\sum_{i=0}^{k-1}|\mu_i|\|u^{n-i}-\bar{u}\|^2 &\geq -\frac{\ell_f}{2}\sum_{q=0}^{k-1}|\mu_q|\sum_{i=0}^{k-1}\sum_{j=0}^{k-1}|\mu_i|\,|\mu_j|\,\|u^{n-i}-u^{n-j}\|^2 \\
&= -\,\ell_f\sum_{q=0}^{k-1}|\mu_q|\sum_{0\leq i<j\leq k-1}|\mu_i|\,|\mu_j|\,\|u^{n-i}-u^{n-j}\|^2.
\end{aligned}
\tag{3.53}
$$

Note that $u^{n-i}-u^{n-j}=\sum_{m=i+1}^{j}\delta u^{n+1-m}$. By the Cauchy–Schwarz inequality, we have $\|u^{n-i}-u^{n-j}\|^2\leq(j-i)\sum_{m=i+1}^{j}\|\delta u^{n+1-m}\|^2$. Inserting this into (3.53) gives

$$
\begin{aligned}
-\frac{\ell_f}{2}\sum_{i=0}^{k-1}|\mu_i|\|u^{n-i}-\bar{u}\|^2 &\geq -\,\ell_f\sum_{q=0}^{k-1}|\mu_q|\sum_{0\leq i<j\leq k-1}|\mu_i|\,|\mu_j|\,(j-i)\sum_{m=i+1}^{j}\|\delta u^{n+1-m}\|^2 \\
&= -\ell_f\sum_{m=1}^{k-1}\bar{c}_m\|\delta u^{n+1-m}\|^2,
\end{aligned}
\tag{3.54}
$$

where $\bar{c}_m$ is defined in (3.47). Substituting (3.54) into (3.51), we obtain

$$
\sum_{i=0}^{k-1}\mu_i(F(u^{n-i}),1)\geq -\ell_f\sum_{m=1}^{k-1}\bar{c}_m\|\delta u^{n+1-m}\|^2.
\tag{3.55}
$$

Substituting (3.55) into the definition of E_G^n in (3.3), and using (3.6), we have

$$
\begin{aligned}
E_G^n &= -\frac{1}{\tau}\boldsymbol{v}_n^\top G_a(\mathcal{M}^{-1}\boldsymbol{v}_n)+\boldsymbol{u}_n^\top G_B(\mathcal{L}\boldsymbol{u}_n)+\sum_{i=0}^{k-1}\mu_i(F(u^{n-i}),1)+\frac{\ell_f}{2}\sum_{i=1}^{k-1}\hat{c}_i\|\delta u^{n+1-i}\|^2 \\
&\geq \frac{1}{\tau}\lambda_a\|(-\mathcal{M})^{-1/2}\boldsymbol{v}_n\|^2+\lambda_B\|\mathcal{L}^{1/2}\boldsymbol{u}_n\|^2+\ell_f\sum_{i=1}^{k-1}\left(\frac{\hat{c}_i}{2}-\bar{c}_i\right)\|\delta u^{n+1-i}\|^2.
\end{aligned}
\tag{3.56}
$$

If $\hat{c}_i/2\geq\bar{c}_i$ for $i=1,\ldots,k-1$, then $E_G^n\geq 0$ holds.

If $\hat{c}_i/2<\bar{c}_i$ for some i, then $\hat{c}_{\min}/2<\bar{c}_{\max}$ and

$$
\begin{aligned}
E_G^n &\geq \frac{1}{\tau}\lambda_a\|(-\mathcal{M})^{-1/2}\boldsymbol{v}_n\|^2+\lambda_B\|\mathcal{L}^{1/2}\boldsymbol{u}_n\|^2+\ell_f\left(\frac{\hat{c}_{\min}}{2}-\bar{c}_{\max}\right)\sum_{i=1}^{k-1}\|\delta u^{n+1-i}\|^2 \\
&\geq \frac{1}{\tau}\lambda_a\|(-\mathcal{M})^{-1/2}\boldsymbol{v}_n\|^2+\frac{\lambda_B}{4}\|\mathcal{L}^{1/2}\boldsymbol{v}_n\|^2+\ell_f\left(\frac{\hat{c}_{\min}}{2}-\bar{c}_{\max}\right)\|\boldsymbol{v}_n\|^2,
\end{aligned}
\tag{3.57}
$$

where we use

$$
\begin{aligned}
\|\mathcal{L}^{1/2}\boldsymbol{v}_n\|^2 &= \sum_{i=0}^{k-2}\|\mathcal{L}^{1/2}\delta u^{n-i}\|^2 \\
&= \sum_{i=0}^{k-2}\Big[\|\mathcal{L}^{1/2}u^{n-i}\|^2+\|\mathcal{L}^{1/2}u^{n-1-i}\|^2-2(\mathcal{L}^{1/2}u^{n-i},\mathcal{L}^{1/2}u^{n-1-i})\Big] \\
&\leq 2\sum_{i=0}^{k-2}\Big[\|\mathcal{L}^{1/2}u^{n-i}\|^2+\|\mathcal{L}^{1/2}u^{n-1-i}\|^2\Big]\leq 4\|\mathcal{L}^{1/2}\boldsymbol{u}_n\|^2.
\end{aligned}
$$

In the case of $\eta = 1$, (2.3) gives $\|\delta u^{n+1-i}\| \le \zeta\|(-\mathcal{M})^{-1/2}\delta u^{n+1-i}\|$. Then $\|\boldsymbol{v}_n\| \le \zeta\|(-\mathcal{M})^{-1/2}\boldsymbol{v}_n\|$. It is straightforward to verify that $E_G^n \ge 0$ when τ satisfies (3.48). In the case of $0 < \eta < 1$, we replace δu^{n+1} by δu^{n+1-i} in (3.45) and then obtain

$$\|\delta u^{n+1-i}\|^2 \le \zeta^2 \left(\eta\xi^{-\frac{1-\eta}{\eta}}\|(-\mathcal{M})^{-1/2}\delta u^{n+1-i}\|^2 + (1-\eta)\xi\|\mathcal{L}^{1/2}\delta u^{n+1-i}\|^2\right) \tag{3.58}$$

for any $\xi > 0$. Then

$$\|\boldsymbol{v}_n\|^2 \le \zeta^2 \left(\eta\xi^{-\frac{1-\eta}{\eta}}\|(-\mathcal{M})^{-1/2}\boldsymbol{v}_n\|^2 + (1-\eta)\xi\|\mathcal{L}^{1/2}\boldsymbol{v}_n\|^2\right). \tag{3.59}$$

To ensure that (3.57) is lower bounded by 0, we then impose

$$\frac{\lambda_a}{\tau} + \ell_f\left(\frac{\hat{c}_{\min}}{2} - \bar{c}_{\max}\right)\zeta^2\eta\xi^{-\frac{1-\eta}{\eta}} \ge 0 \quad \text{and} \quad \frac{\lambda_B}{4} + \ell_f\left(\frac{\hat{c}_{\min}}{2} - \bar{c}_{\max}\right)\zeta^2(1-\eta)\xi \ge 0. \tag{3.60}$$

By taking

$$\xi = \frac{\lambda_B}{4\ell_f(\bar{c}_{\max} - \hat{c}_{\min}/2)\zeta^2(1-\eta)},$$

the second inequality of (3.60) holds naturally and we then obtain the time-step restriction (3.48). ∎

We present the property of G_B to obtain the consistency of E_G^n with $E[u(t_n)]$.

LEMMA 3.5 (Consistency). *The matrix G_B in Lemma 3.3 satisfies*

$$\mathbf{1}_k^\top G_B \mathbf{1}_k = \frac{1}{2}\sum_{i=0}^{k-1}\mu_i, \quad \textit{where} \quad \mathbf{1}_k = [1,\dots,1]^\top \in \mathbb{R}^k. \tag{3.61}$$

Proof. Multiplying (3.30) from the left by $\mathbf{1}_{k+1}^\top$ and right by $\mathbf{1}_{k+1}$ yields

$$\mathbf{1}_{k+1}^\top U_B \mathbf{1}_{k+1} = \frac{1}{2}\mathbf{1}_{k+1}^\top\left(\boldsymbol{\nu}(\boldsymbol{B}^{(k)})^\top + \boldsymbol{B}^{(k)}\boldsymbol{\nu}^\top\right)\mathbf{1}_{k+1} - \mathbf{1}_k^\top G_B\mathbf{1}_k + \mathbf{1}_k^\top G_B \mathbf{1}_k = 0,$$

where we have used the property $\mathbf{1}_{k+1}^\top\boldsymbol{\nu} = 0$ that follows from the definition (3.1). Since U_B is positive semi-definite, it follows that $U_B\mathbf{1}_{k+1} = \mathbf{0}$. Then multiplying (3.30) from the right by $\mathbf{1}_{k+1}$ yields

$$U_B\mathbf{1}_{k+1} = \frac{1}{2}\boldsymbol{\nu}(\boldsymbol{B}^{(k)})^\top\mathbf{1}_{k+1} - \begin{bmatrix}\boldsymbol{g}\\0\end{bmatrix} + \begin{bmatrix}0\\\boldsymbol{g}\end{bmatrix} = \mathbf{0}, \tag{3.62}$$

where $\boldsymbol{g} := G_B\mathbf{1}_k \in \mathbb{R}^k$. Writing $\boldsymbol{g} = [g_1,\dots,g_k]^\top$, by (2.6), we have $(\boldsymbol{B}^{(k)})^\top\mathbf{1}_{k+1} = 1$, and obtain from (3.62) the elementwise recurrence: $g_1 = \nu_0/2 = \mu_0/2$ and $g_{j+1} - g_j = \nu_j/2$ for $1 \le j \le k-1$. By summing this recurrence, we readily obtain

$$g_i = \frac{1}{2}\sum_{j=0}^{i-1}\nu_j = \frac{\mu_{i-1}}{2} \quad \text{for} \quad i = 1,\dots,k.$$

That is, $\boldsymbol{g} = \frac{1}{2}[\mu_0,\mu_1,\dots,\mu_{k-1}]^\top$. Therefore $\mathbf{1}_k^\top G_B\mathbf{1}_k = \mathbf{1}_k^\top\boldsymbol{g} = \frac{1}{2}\sum_{i=0}^{k-1}\mu_i$. ∎

Remark 3.6. *Assume that $u(t)$ is sufficiently smooth in time, and let $u^{n-i} = u(t_{n-i})$. Then $\delta u^{n-j} = u(t_{n-j}) - u(t_{n-j-1}) = \mathcal{O}(\tau)$ for $j = 0, \dots, k-2$, so that $\boldsymbol{v}_n = \mathcal{O}(\tau)$. It follows from (3.3) that*

$$E_G^n = \boldsymbol{u}_n^\top G_B(\mathcal{L}\boldsymbol{u}_n) + \sum_{i=0}^{k-1} \mu_i(F(u^{n-i}), 1) + \mathcal{O}(\tau).$$

As $\tau \to 0$, we have $\boldsymbol{u}_n \to u(t_n)\mathbf{1}_k$ and $u^{n-i} \to u(t_n)$. Hence, by Lemma 3.5,

$$E_G^n \to (\mathbf{1}_k^\top G_B \mathbf{1}_k)(u(t_n), \mathcal{L}u(t_n)) + \sum_{i=0}^{k-1} \mu_i(F(u(t_n)), 1) = E[u(t_n)].$$

Thus, the modified energy E_G^n is consistent with the continuous energy $E[u]$.

3.3. Proof of Theorem 3.1. We verify (3.28) in Lemma 3.3 by applying Proposition 3.2 to U_a and U_B.

For U_a, let $\rho(z) = M(z; \boldsymbol{a}^{(k)})$, $\sigma(z) = M(z; \boldsymbol{\mu})$, $\lambda(z) = z^{k-1}$, and $\gamma = \alpha$. By the assumptions of the theorem, condition (iii) of Proposition 3.2 holds. Hence, by (iii) $\iff$ (v), there exist a symmetric positive definite matrix G and a vector $\boldsymbol{q}$ such that (3.11) is satisfied. Setting $G_a = G/2$ and substituting (3.11) into (3.29), we obtain $U_a = \frac{1}{2}\boldsymbol{q}\boldsymbol{q}^\top + \alpha \boldsymbol{e}_1 \boldsymbol{e}_1^\top$, so the U_a-condition in (3.28) follows.

For U_B, let $\rho(z) = M(z; \boldsymbol{B}^{(k)})$, $\sigma(z) = M(z; \boldsymbol{\nu}) = (z-1)M(z; \boldsymbol{\mu})$, $\lambda(z) = z^k - z^{k-1}$, and $\gamma = \beta$. We verify condition (iii) of Proposition 3.2. Condition (iii)(a) follows from $M(z; \boldsymbol{\mu}) \neq 0$ for $|z| > 1$. On $|z| = 1$, writing $z = e^{\mathrm{i}\theta}$ gives

$$|\lambda(z)|^2 = |z^k - z^{k-1}|^2 = |z-1|^2 = 2(1 - \cos\theta),$$

and thus the assumed inequality is condition (iii)(b). It remains to check (iii)(c). If $M(z; \boldsymbol{\mu})$ had a root z_* on $|z| = 1$, then $\sigma(z_*) = 0$, and (iii)(b) would imply

$$0 = \mathrm{Re}\{\rho(z_*)\sigma(\bar{z}_*)\} \geq \beta|\lambda(z_*)|^2.$$

Since $\beta > 0$, this yields $\lambda(z_*) = 0$, hence $z_* = 1$. Therefore, the only root of σ on $|z| = 1$ is $z_0 = 1$. Moreover, $\sigma(z) = (z-1)M(z; \boldsymbol{\mu})$, and $M(1; \boldsymbol{\mu}) = \sum_{i=0}^{k-1} \mu_i = 1$ by (1.4), so $z_0 = 1$ is simple and $\sigma'(1) = M(1; \boldsymbol{\mu}) = 1$. By (2.6), we have $\rho(1) = M(1; \boldsymbol{B}^{(k)}) = 1$, and hence $\frac{\rho(1)}{\sigma'(1)}\,\overline{1} = 1 > 0$. Thus condition (iii)(c) also holds.

Applying again (iii) $\iff$ (v), we obtain a symmetric positive definite matrix G and a vector $\boldsymbol{q}$ such that (3.11) holds. Setting $G_B = G/2$ and substituting (3.11) into (3.30), we get

$$U_B = \frac{1}{2}\boldsymbol{q}\boldsymbol{q}^\top + \beta\boldsymbol{\lambda}\boldsymbol{\lambda}^\top, \quad \text{where } \boldsymbol{\lambda} = [1, -1, 0, \dots, 0]^\top \in \mathbb{R}^{k+1},$$

which yields the U_B-condition in (3.28).

Therefore all assumptions of Lemma 3.3 are satisfied, and the conclusion $E_G^{n+1} \leq E_G^n$ under the time-step restriction (3.7) follows from Lemma 3.3. ∎

REFERENCES

[1] C. Baiocchi and M. Crouzeix, *On the equivalence of A-stability and G-stability*, Appl. Numer. Math., 5 (1989), pp. 19–22, https://doi.org/10.1016/0168-9274(89)90020-2.

[2] H. Brezis, *Functional analysis, Sobolev spaces and partial differential equations*, Universitext, Springer, New York, 2011.
[3] G. G. Dahlquist, *G-stability is equivalent to A-stability*, BIT, 18 (1978), pp. 384–401.
[4] I. Daubechies, *Ten lectures on wavelets*, vol. 61 of CBMS-NSF Regional Conference Series in Applied Mathematics, Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 1992.
[5] Z. Fu, T. Tang, and J. Yang, *Energy diminishing implicit-explicit Runge-Kutta methods for gradient flows*, Math. Comp., 93 (2024), pp. 2745–2767.
[6] E. Hairer and G. Wanner, *Solving ordinary differential equations. II*, vol. 14 of Springer Series in Computational Mathematics, Springer-Verlag, Berlin, revised ed., 2010. Stiff and differential-algebraic problems.
[7] R. A. Horn and C. R. Johnson, *Matrix analysis*, Cambridge university press, 2012.
[8] X. Li, Z. Qiao, and C. Wang, *Convergence analysis for a stabilized linear semi-implicit numerical scheme for the nonlocal Cahn-Hilliard equation*, Math. Comp., 90 (2021), pp. 171–188.
[9] J. Shen and X. Yang, *Numerical approximations of Allen-Cahn and Cahn-Hilliard equations*, Discrete Contin. Dyn. Syst., 28 (2010), pp. 1669–1691.